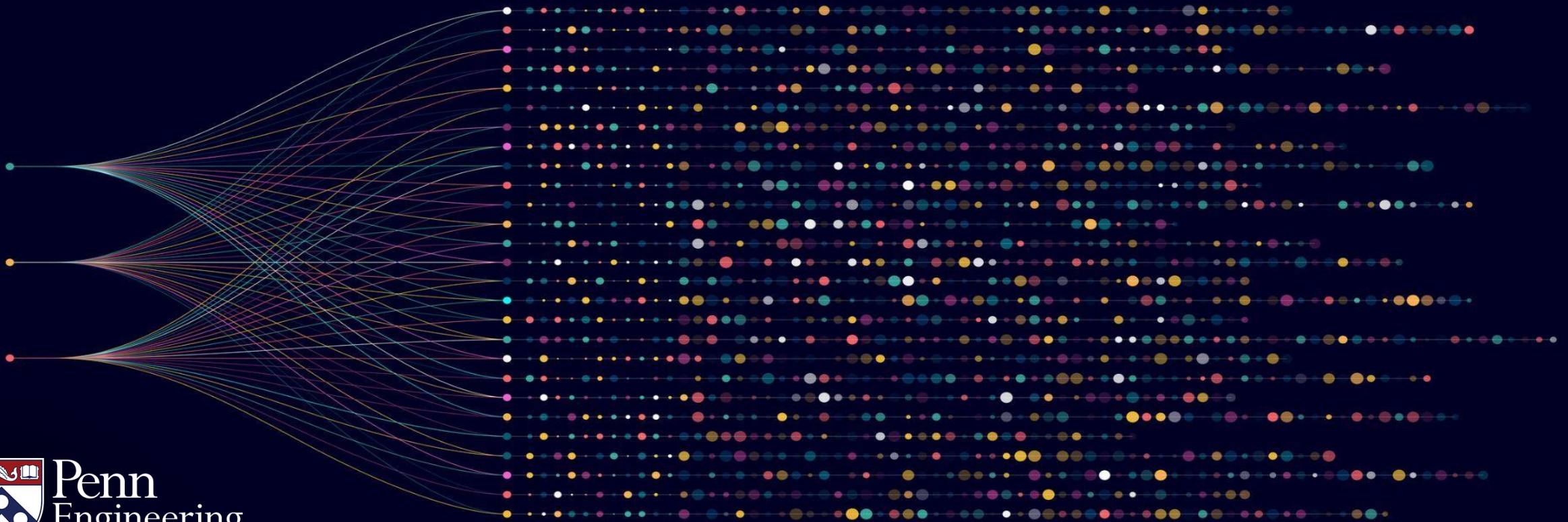


Algorithms for Big Data

Anindya De

Erik Waingarten

Small Summaries from Sampling

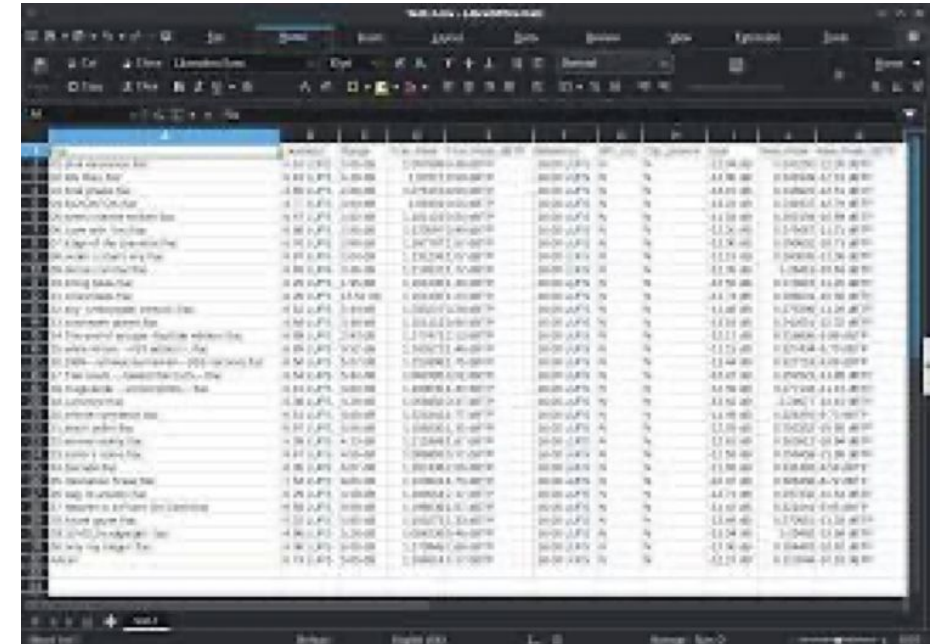


Objectives: Computing Large Averages

- Application: small summaries via random sampling
- Small failure probability critically important

Motivating Scenario:

- Users have many attributes
 - approx. statistics on a lot of different user behavior
- How many European? How many are German? British? How many posted today? Yesterday? Logged in twice in a week? ...
- Unclear which questions will arise



The image shows a screenshot of a large data table, likely from a database or a data analysis tool. The table has many columns and rows, with the first column containing user names. The data appears to be organized into groups, possibly by user type or activity. The table is displayed in a window with a standard toolbar at the top.

Rows are users,
columns are attributes

Motivating Scenario:

For each question:

Algorithm



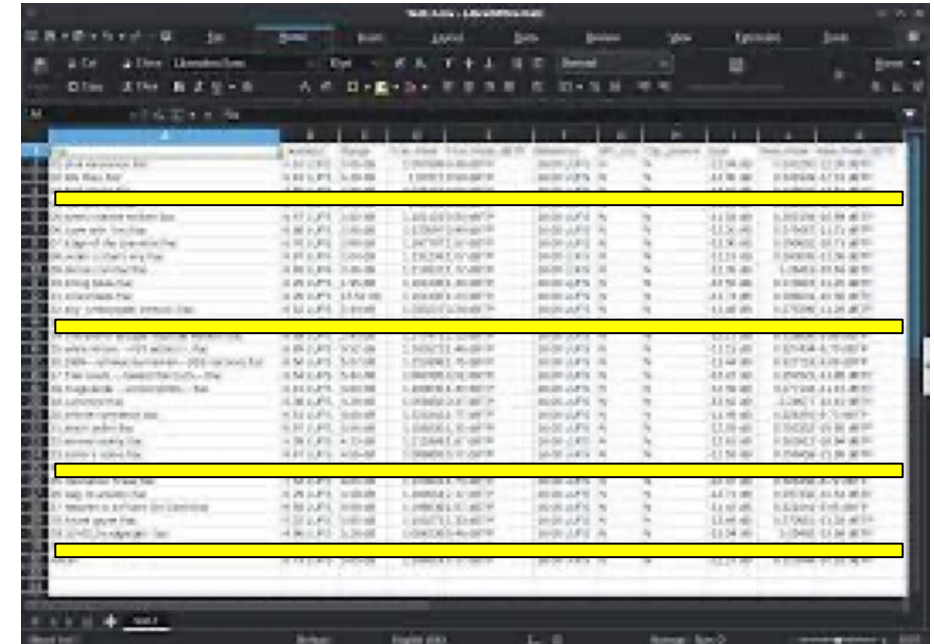
↓
 $\hat{S}$

The image shows a screenshot of a data table with many rows and columns. The rows represent individual users, and the columns represent various attributes or features. The table is displayed in a software interface, likely a spreadsheet or database viewer, with a dark theme. The columns are labeled with various attributes, and the rows contain numerical values for each attribute.

Rows are users,
columns are attributes

Motivating Scenario:

- Each execution generates a random sample of users
- If repeated for M questions, would be using M times q queries

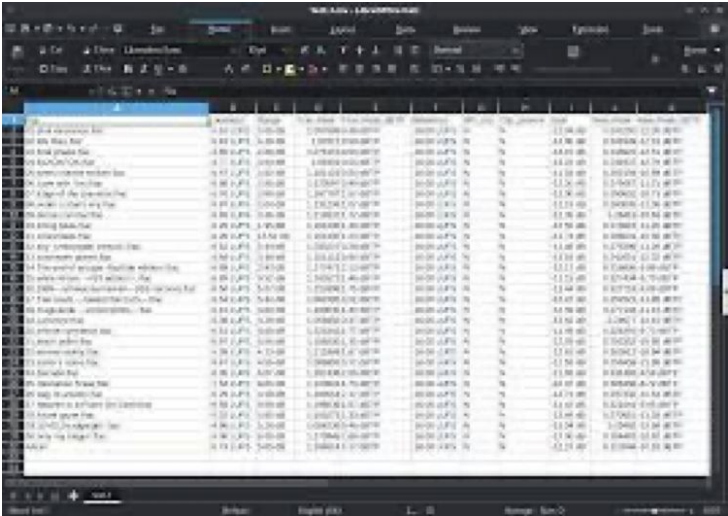


The image shows a screenshot of a data table, likely from a database or spreadsheet application. The table has multiple columns and rows. Several rows are highlighted in yellow, indicating a selection of data. The columns appear to contain numerical and categorical data, representing attributes of users. The rows represent individual users, with each row containing a unique set of attribute values.

Rows are users,
columns are attributes

Motivating Scenario: Is it possible to...?

- Generate a *single* set of samples
- For each question, use the *same* set of samples to execute the algorithm



A screenshot of a data table, likely from a spreadsheet application. The table has many columns and rows, with some cells highlighted in blue. The data appears to be numerical and categorical, possibly representing experimental results or simulation data.

$$\mathbf{i}_1, \dots, \mathbf{i}_q \sim [n]$$



q



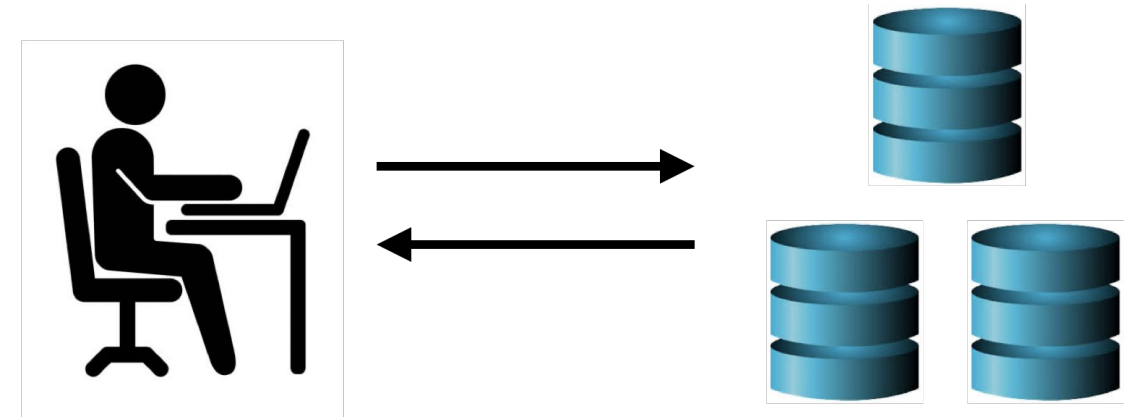
A smaller screenshot of a data table, similar to the one above, representing a subset of the data. It shows a few rows and columns of data, with some cells highlighted in blue.

Benefits:

- Ideally, fewer samples than one batch per question
- Samples can be pre-generated and used multiple times
- Fewer interactions with central server (also be preemptively done)

Mathematical Formalism:

- Domain $\mathcal{X}$ of possible objects
 - ex. possible space of users
- There is a dataset $P \subset \mathcal{X}$ of n objects
 - ex. possible users
- Set of functions $\mathcal{F}$ mapping domain to $[0,1]$



$$f \in \mathcal{F} \quad : \quad f: \mathcal{X} \rightarrow [0, 1]$$

Example: Country of Origin for Users

- All possible users $\mathcal{X}$.
- Our dataset of users $P \subset \mathcal{X}$.
- Collection of 195 functions:

$$\mathcal{F} = \{f_{\text{USA}}, f_{\text{Canada}}, f_{\text{Egypt}}, \dots\}$$

$$f_{\text{USA}}(x)$$

$$= \begin{cases} 1 & x \text{ from USA} \\ 0 & \text{otherwise} \end{cases}$$

Main Result:

Theorem: For any collection of functions $\mathcal{F}$, any dataset P , and any accuracy parameter ϵ and failure probability δ .

Letting

$$q = \frac{\log(2|\mathcal{F}|/\delta)}{2\epsilon^2}$$

A summary of q random samples can ϵ -approximate every function $f \in \mathcal{F}$ with probability at least $1 - \delta$

Applying the Theorem:

- Fix collection of functions $\mathcal{F}$
- Estimate statistic:

$$\frac{1}{q} \sum_{t=1}^q f(x_{i_t})$$

- Number of samples:

$$O(\log(|\mathcal{F}|/\delta)/\epsilon^2)$$

A screenshot of a spreadsheet application showing a large table of data. The table has many columns and rows, with some cells highlighted in blue. The data appears to be numerical values, possibly representing function outputs or statistical data.

$$i_1, \dots, i_q \sim [n]$$



q

A smaller screenshot of a spreadsheet showing a subset of the data from the top image. The variable q is highlighted in the first column of the table.

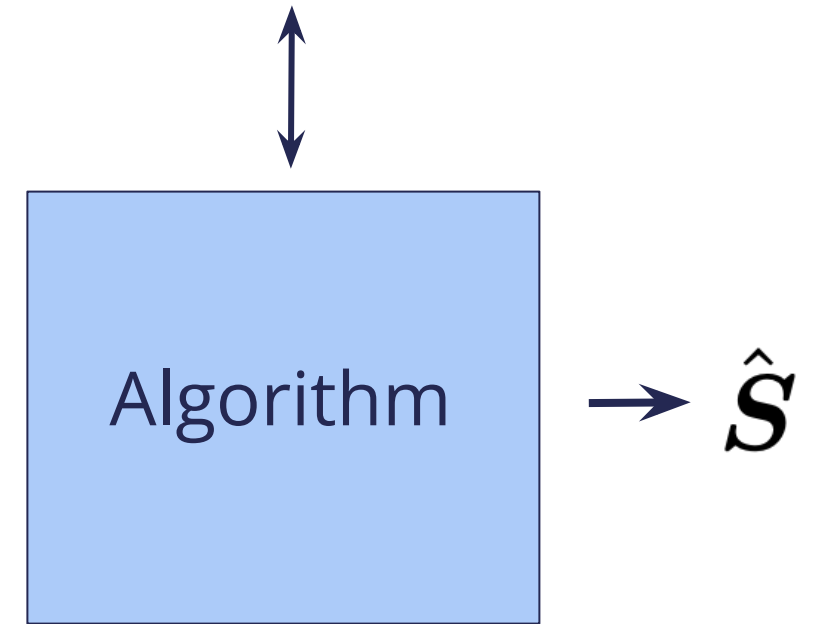
Analysis: Union bound over collection.

- Consider single statistic



$$f: \mathcal{X} \rightarrow [0, 1]$$

$$\frac{1}{n} \sum_{t=1}^n f(x_i) \stackrel{?}{=} \frac{1}{q} \sum_{t=1}^q f(x_{i_t})$$



- Failure probability:

$$\Pr \left[|S - \hat{S}| \geq \epsilon \right] \leq 2e^{-2\epsilon^2 q}$$

Analysis: Union bound over collection.

- Failure probability:

$$\Pr \left[|S - \hat{S}| \geq \epsilon \right] \leq 2e^{-2\epsilon^2 q}$$

- When $q = \frac{\log(2|\mathcal{F}|/\delta)}{2\epsilon^2}$,

$$\leq \delta/|\mathcal{F}|$$

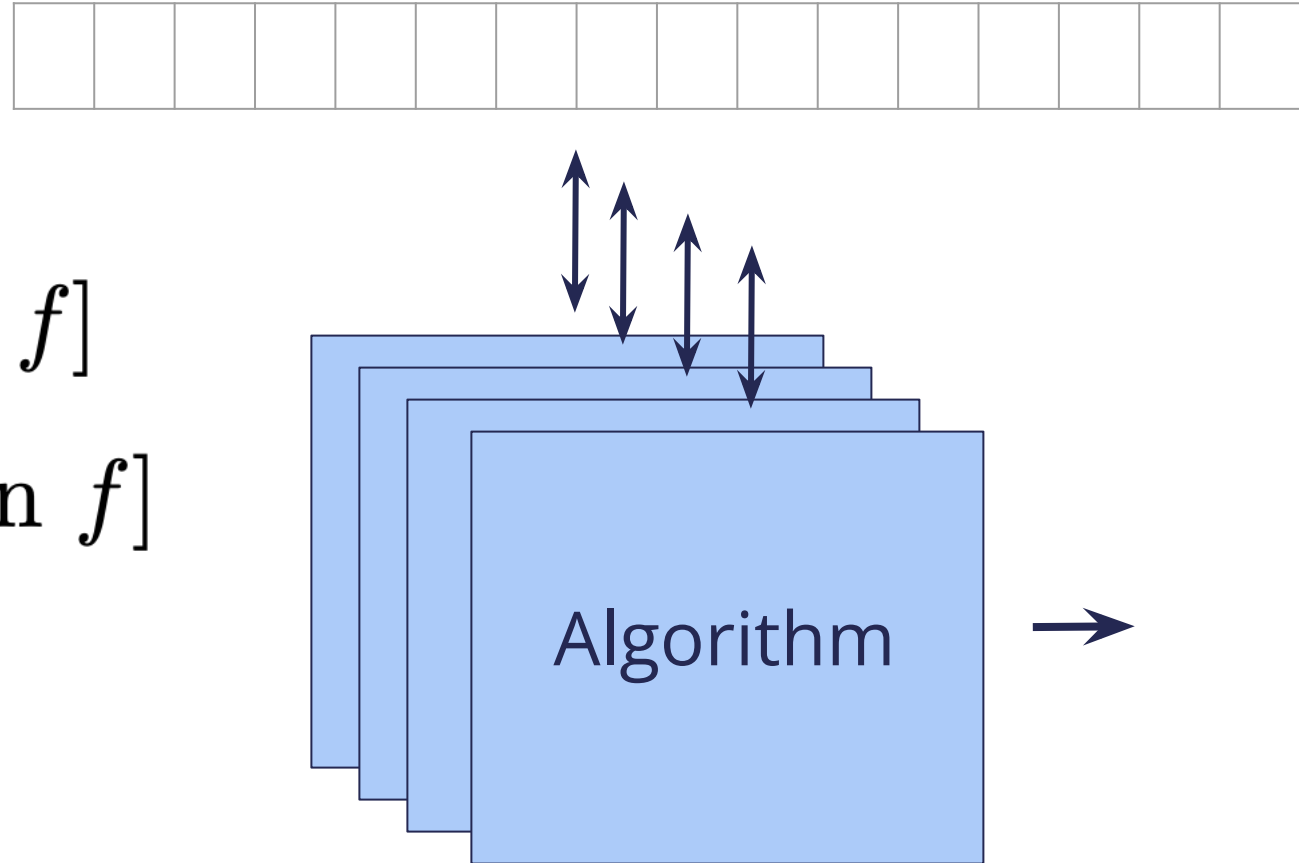


Analysis: Union bound over collection.

- Probability that there exists a mistake:

$$\begin{aligned}\Pr [\exists f \in \mathcal{F} : \text{mistake on } f] \\ &\leq \sum_{f \in \mathcal{F}} \Pr [\text{mistake on } f] \\ &\leq |\mathcal{F}| \cdot \delta / |\mathcal{F}|\end{aligned}$$

by union bound



Summary

- Random sampling gives small summaries (“sketches”) of big data
 - For fixed collection of statistics, dramatic data reduction

		ϵ		
δ		0.1	0.01	0.001
	0.1	413	41K	4.1M
	0.01	529	53K	5.3M
	0.001	644	64K	6.4M

- Important caveat: functions must be fixed in advanced!